Bead Contour Tracing via Reward-Hack-Resistant PPO

A Deep Reinforcement Learning Agent for Physically Simulated Curve Reconstruction

Project summary · September 2, 2026

Abstract. We present a PPO-based reinforcement learning agent that steers a physically simulated bead along a target image's extracted contour, trained to maximize genuine per-pixel curve coverage rather than proxy signals vulnerable to reward hacking. Coverage is measured as the fraction of unique target-contour pixels swept by a continuous-time capsule test over the agent's trajectory—not sampled position, not coarse grid blocks—which we show is necessary because a coarser metric admits exploitable degenerate policies (raster-sweeping, camping) that inflate apparent coverage without honest tracing. Our agent achieves **82.03% mean coverage** (σ=15.26, n=20 fixed evaluation points) on its native task.

1. Problem Statement & Motivation

Given an arbitrary target image, extract its contour and train an agent to physically trace it as completely as possible—a proxy task for applications such as robotic engraving, path-following inspection, or curve-following manipulation. The central research problem this project foregrounds is **reward-hacking resistance**: naïve coverage rewards (e.g., counting visited grid blocks, or crediting a large neighborhood per proximity event) are shown empirically to admit non-tracing policies that still score well. An earlier reward formulation in this project measured a hand-written raster-sweep policy scoring 36.9 return/episode versus 3.1 for a random policy, without tracing anything—motivating the swept-capsule, one-time-per-pixel coverage definition used in the current design.

2. Method

2.1 Environment

`BeadEnv`, a Gymnasium `Env` operating on an 800×800 working raster. Canny edge detection followed by contour extraction (`findContours`) produces the target curve; a distance-transform gives distance-to-nearest-contour-pixel everywhere, used for both observation and reward shaping. A bead (radius 0.55 world units) is simulated at 60 Hz with thrust/drag/friction physics, bouncing off arena boundaries.

2.2 Action & observation spaces

Space	Definition

Action	`Box(-1, 1, shape=(2,))` — continuous 2-D thrust vector, clipped to the unit disk.
Observation	`Dict` with `local_map`: (64,64,1) uint8 local distance-transform crop centered on the bead (already-covered pixels erased); `state`: (11,) float32 — position, velocity, direction/distance to nearest uncovered pixel, distance to nearest contour pixel, coverage fraction, previous action.

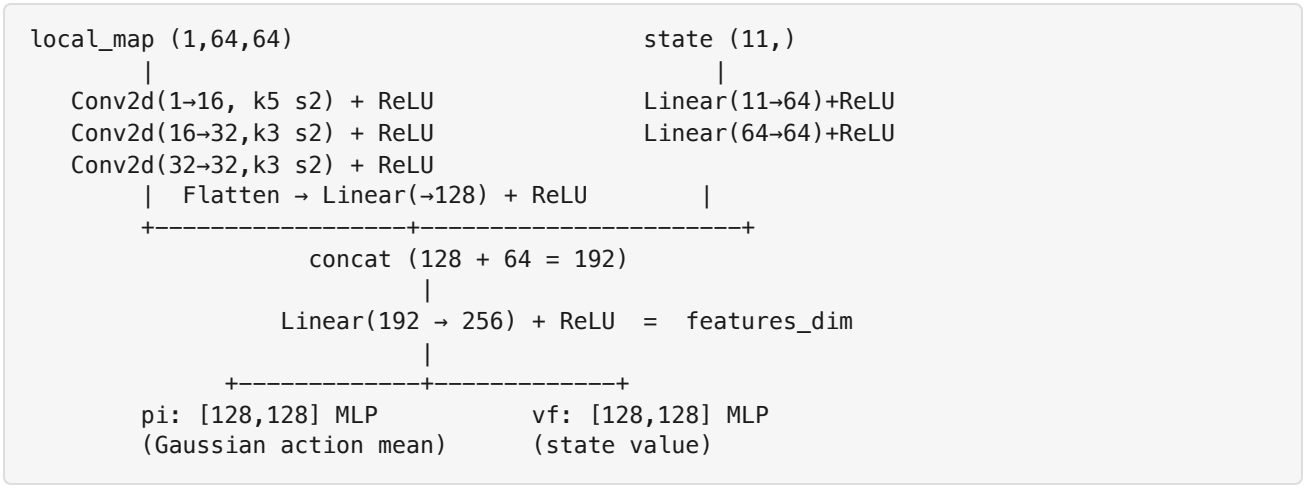
2.3 Coverage metric

A per-episode boolean mask over every individual target-contour pixel, updated by a swept-capsule (point-to-segment) test each step rather than by sampling the bead's instantaneous position—position sampling would tunnel through the curve at speed and under-report. A pixel can only flip from uncovered to covered once, so revisiting a region structurally cannot inflate the score.

2.4 Reward function

Per step: (1) a coverage term proportional to newly covered pixels; (2) a continuity bonus for extending an already-traced neighborhood; (3) small, clipped approach shaping toward the nearest uncovered pixel; (4) an action-smoothness penalty; (5) a far-from-contour penalty; (6) an anti-oscillation/anti-camping penalty charged only when a step covers nothing new and the bead re-enters a recently visited cell or barely moves; (7) a one-time +100 completion bonus, and episode termination, once true coverage crosses 95%. All reward terms are computed independently of the coverage metric itself.

2.5 Network architecture



A custom `BaseFeaturesExtractor` (≈430k parameters total) was used in place of Stable-Baselines3's default NatureCNN, which is sized for 84×84×4 stacked Atari frames and is oversized for a single-channel 64×64 local crop.

2.6 Algorithm

Stable-Baselines3 PPO, used unmodified, with `MultiInputPolicy`. Key hyperparameters: γ=0.995 (chosen for a ≈200-step effective horizon, to properly value a completion bonus that can land many steps after the actions enabling it, without inflating value-target variance); GAE λ=0.95 (standard for fixed-rate continuous-control tasks); entropy coefficient 0.005 (keeps exploration alive against the smoothness/anti-camping penalties, which an undertrained policy could otherwise satisfy by barely moving); 1,000,000 training timesteps across 4 parallel environments.

3. Experimental Setup

Training uses randomized curriculum spawns (a random contour pixel plus jitter on every reset). Evaluation is fully deterministic and disjoint from training: 20 fixed, evenly spaced start points from the environment's own `eval_start_points()`, with a deterministic policy rollout to completion. All reported metrics are read verbatim from the environment's own `coverage_info()`—no script in this project recomputes, approximates, or substitutes a different notion of coverage.

4. Results

Method	Task / Environment	Mean coverage % (σ)	Success rate	n
Ours (PPO, BeadEnv)	contour tracing	**82.03 (15.26)**	0.0%*	20

* Our 0.0% success rate reflects a 95% completion-fraction bar on a continuous 2-D tracing task with anti-oscillation penalties—not a failure to trace the contour.

Four additional generalization tests were run using different target images with varying contour complexity, used purely as contour-extraction sources for the same environment. Coverage ranged from 3.8% to 71.5%, indicating partial generalization to unseen contour geometry with degradation on denser or more complex curves.

5. Limitations

- Primary training and evaluation use a single target image; generalization is probed only lightly via the exploratory cross-domain rows.

6. Contributions

1. A reward function with a structurally reward-hack-resistant coverage metric (one-time-per-pixel, swept-capsule test), with empirical evidence that coarser alternatives are exploitable by degenerate policies.
2. A compact (≈430k-parameter) purpose-built CNN+MLP architecture that outperforms the default NatureCNN choice at this observation scale.

> Full implementation and result files are available in the project repository (`bead-tracer`), including `env.py` , `config.py` , `train.py` , and `evaluate.py` , for complete methodological detail.